Scaling a database vertically is the easier option, but it gets expensive. High-end hardware is more expensive than commodity hardware. Besides that, at some point, you will not find more powerful machines on the market anymore.

Scaling a traditional relational database horizontally is difficult because transactional guarantees such as atomicity, consistency, isolation, and durability—also known as *ACID*—require communication among all nodes of the database during a two-phase commit. To see what we're talking about, here's how a simplified two-phase commit with two nodes works, as illustrated in figure 12.1:

1. A query is sent to the database cluster that wants to change data (`INSERT`, `UPDATE`, `DELETE`).
2. The database transaction coordinator sends a commit request to the two nodes.
3. Node 1 checks whether the query could be executed. The decision is sent back to the coordinator. If the nodes decide yes, the query can be executed, it must fulfill this promise. There is no way back.
4. Node 2 checks whether the query could be executed. The decision is sent back to the coordinator.
5. The coordinator receives all decisions. If all nodes decide that the query could be executed, the coordinator instructs the nodes to finally commit.
6. Nodes 1 and 2 finally change the data. At this point, the nodes must fulfill the request. This step must not fail.